Session 4: Sampling Errors - Alpha & Beta Analysis

Context & Objective

Textbook: Chapter 9 – Hypothesis Testing & Sampling Errors

Objective: Students build an error analyzer skill to calculate the probability of a Type II error (beta) and the power of the test, evaluating customer risk during quality inspections.

1. Antigravity Skill Definition (type2-error-analyzer)

Skill Name: type2-error-analyzer

Purpose: Calculates the probability of Type II error (β) and the power (1 - β) of a hypothesis test, allowing managers to evaluate the risk of failing to detect a true effect.

1.1 Core Logic & Formulas

Step 1: Input Parameters

- μ_0 (mu0): Baseline/null hypothesis mean
- μ_1 (mu1): Hypothesized/alternative mean (the effect to detect)
- σ (sigma): Population standard deviation
- n: Sample size
- α (alpha): Significance level (e.g., 0.05)

Step 2: Calculate Critical Mean Under H_0 (for upper-tailed test)

$$x_crit = \mu_0 + z_\alpha \times (\sigma / \sqrt{n})$$

Step 3: Standardize Critical Mean Under H_1

$$z = (x_crit - \mu_1) / (\sigma / \sqrt{n})$$

Step 4: Calculate Type II Error Probability

$$\beta = P(Z < z) = \Phi(z)$$ [cumulative standard normal probability]

Step 5: Calculate Power of the Test

$$Power = 1 - \beta$$

2. Execution Prompt

Evaluate the Risk of Missing a Calibration Error

- Baseline target weight (mu0) = 16.0 oz.
- actual shifted weight (mu1) = 16.2 oz.
- standard deviation (sigma) = 0.4 oz, sample size (n) = 25, significance level (alpha) = 0.05.

Calculate the critical value x_crit under H0, the probability of a Type II error (beta), and the power of the test

Given Data:

Parameter	Value
Baseline target weight (μ_0)	16.0 oz
Suspected shifted weight (μ_1)	16.2 oz
Standard deviation (σ)	0.4 oz
Sample size (n)	25
Significance level (α)	0.05 (5%)
Test type	Upper-tailed (one-sided)

Assignment Task:

1. Calculate the critical value x_crit under H_0 using the formula above.
2. Compute the standardized z-value for x_crit under H_1 ($\mu_1 = 16.2$).
3. Calculate beta (β) as the cumulative probability $P(Z < z)$.
4. Calculate the power of the test as $1 - \beta$.
5. Interpret the results: What is the probability of missing a 0.2 oz shift at n = 25?

3. Output & Interpretation

Solution for n = 25:

Calculated Metric	Result
Critical mean (x_crit)	16.13 oz
Standardized z-value	−0.84
Type II Error Probability (β)	79.4%
Power of Test ($1 - \beta$)	20.6%

Managerial Interpretation:

Risk Assessment: With a sample size of 25, there is a 79.4% chance that the test will fail to detect the 0.2 oz shift from the target of 16.0 oz to the suspected 16.2 oz. In other words, even if the calibration has truly drifted and the line is overfilling by 0.2 oz, the quality test will miss this 79.4% of the time at n = 25. This is an unacceptably high risk for quality control.

The quality manager decides to increase the sample size to n = 100 to improve the power of the test.

Revised Results at n = 100:

Metric	At n = 100
Type II Error (β)	11.5%

Power of Test	88.5%

Improved Outcome: By increasing the sample size from 25 to 100, the power jumps to 88.5%, meaning the quality test now has an 88.5% chance of correctly detecting a true 0.2 oz shift. The Type II error probability falls to just 11.5%, a 7-fold reduction from the original n = 25 scenario.

Decision: The manager approves the increase to n = 100 because 88.5% power is acceptable for process control. The line will now be monitored with improved sensitivity to detect calibration drift, reducing the risk of undetected product quality issues.

4. Inferences

- Beta (β) is not fixed; it depends on the effect size and sample size.
- Power increases with larger sample sizes — bigger samples reduce the chance of missing a real effect.
- The trade-off between Type I (α) and Type II (β) errors is a core principle: reducing β typically requires increasing n.
- Managers use power analysis to decide whether a proposed sample size is sufficient for their business tolerance of risk.

